# Probabilistic Machine Learning

## Lecture 01: Introduction

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Speaker notes

The lecture argues from a single prediction to a course programme. A class label hides how sure a model is; a probability attached to it supports ranking, cost-based thresholds and deferral. Uncertainty has two sources — variation in the outcome given the observed features (aleatoric) and limited knowledge of the model (epistemic) — and it enters every component of a learning system: data, model family, parameters, objective, optimisation and evaluation. A learned prediction is an inductive or abductive inference rather than a deduction, so it needs a calculus of degrees of support; probability is that calculus for representing, combining and updating beliefs, and decisions add costs. The formal vocabulary (joint, conditioning, inference) is introduced in Lecture 02; this lecture uses it only by name.

# Why Probability?

**Guiding Question:** Two models both say "fraud". What would you need to know before acting on either — and why does a label not tell you?

# Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

| | | | |
|---|---|---|---|
| Café Central<br>€4.20<br>legitimate | Supermarket<br>€63.10<br>legitimate | **Electronics Online**<br>€1,299.00<br>**fraud** | Petrol Station<br>€78.40<br>legitimate |
| **Gift Cards ×12**<br>€600.00<br>**fraud** | Bookshop<br>€24.90<br>legitimate | **Hotel Booking**<br>€410.00<br>**fraud** | Pharmacy<br>€12.75<br>legitimate |

Speaker notes

The batch holds eight card transactions. Step 1: the classifier labels each one — five legitimate, three fraud. Step 2: the legitimate ones recede, leaving the three flagged cases: a €1,299 online electronics order, twelve gift cards for €600, and a €410 hotel booking. These are different situations, yet the model's output for all three is the same word.

The three questions have factual answers. The cases need not be equally suspicious, but a label cannot show it: most hard classifiers threshold an internal score, and the threshold discards everything except which side of it a case falls on. With labels only, all three cases must receive the same treatment. Treating them differently requires a graded quantity — ideally the probability of fraud given the features — together with the cost of each kind of error.

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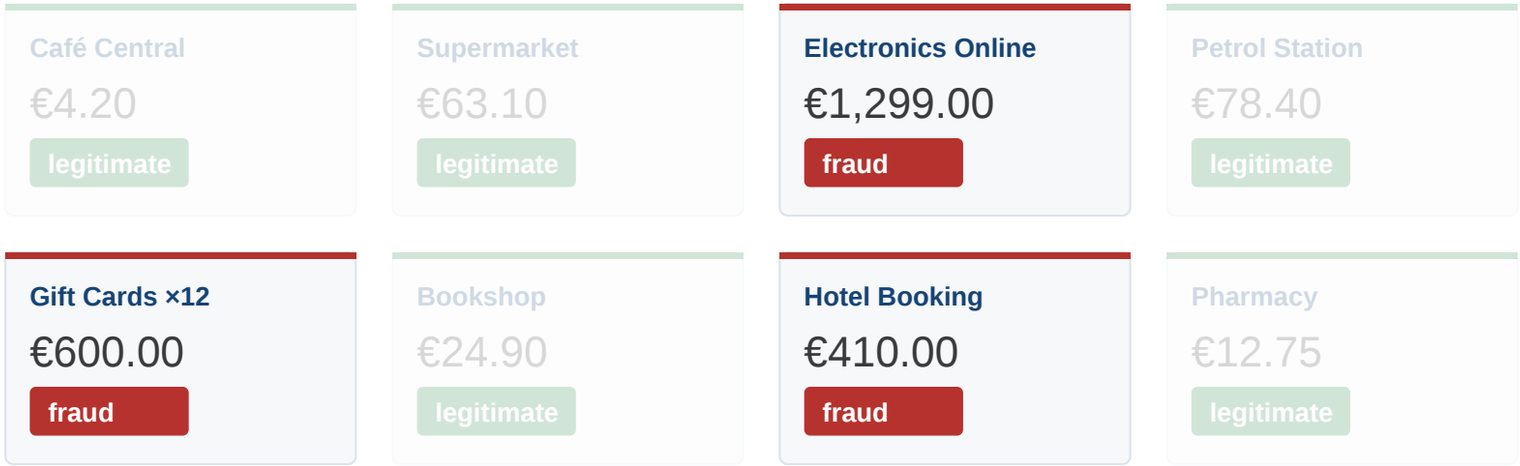
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Are the three flagged transactions equally suspicious?

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Would you block all three cards?

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Are the three flagged transactions equally suspicious?

Would you block all three cards?

What would you need to know to treat them differently?



# The Same Batch, With a Number Attached




What can you do now that you could not do a minute ago?

Speaker notes

Same batch, same model. Step 1 labels the batch again; step 2 attaches the model's probability of fraud and a bar to each flagged case: 0.98 for the electronics order, 0.55 for the gift cards, 0.51 for the hotel booking. The three cases that were indistinguishable now separate.

The number supports actions a label cannot: ranking the cases, acting at once on the near-certain one, routing the two borderline ones to a person, and asking why the model is unsure about them. It is not a more precise label but a different kind of information.

A score of 0.55 is a probability in the operational sense only if the model is *calibrated*: among all cases scored 0.55, about 55% are fraud. Many classifier scores are monotone in the probability but not equal to it — margins, softmax outputs of over-confident networks, or scores from models trained with re-weighted classes. Lecture 04 treats when a model's output earns the name.

# What a Number Lets Us Do

| The number lets us | What that means for this batch |
|---|---|
| **Compare and rank** | separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud" |
| **Weigh consequences** | set the threshold from the cost of a missed fraud against the cost of an annoyed customer |
| **Defer deliberately** | send 0.55 and 0.51 to a human; ask whether more data would move them |
| **Combine and communicate** | merge evidence from several signals; report *how sure* we are, not only what we think |

Speaker notes

The four uses are distinct. *Comparing* requires only a monotone score; *weighing consequences* requires a calibrated probability, because expected costs are computed from it (worked out in the mini-exercise that follows). *Deferring* uses a band of probabilities around the decision threshold in which a human check costs less than the expected cost of either automatic action; whether more data would narrow that band depends on the source of the uncertainty. *Combining* evidence from several signals is principled only for probabilities: under conditional independence of the signals given the class, their likelihood ratios multiply.

Representing uncertainty and reasoning with it are separate requirements. A number alone does not say how to combine it with another number, update it on new evidence, or turn it into an action; those rules are what the course develops.

# What a Number Lets Us Do

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> **We need to model uncertainty** — and we need to be able to **reason with it**.

# From Prediction to Managed Uncertainty

**Predict**
a label or a value

→

**Quantify**
how sure we are

→

**Model**
represent it explicitly

→

**Reason**
combine, compare, update

→

**Manage**
act, defer, or gather data

And once we want to manage it — **where does the uncertainty actually come from?**

> Probabilistic machine learning is not a different way of producing predictions. It is a **language for reasoning about predictions that are not certain**.

Speaker notes

The five stages read as one sentence: a prediction (label or value) is quantified by how sure the model is, that uncertainty is represented explicitly in a model, the representation is used to combine, compare and update, and the result is managed — acted on, deferred, or reduced by gathering data.

The right way to manage uncertainty depends on its cause. Gathering data reduces uncertainty that stems from limited knowledge; it does not reduce variation inherent in the outcome given the features.

# Mini-Exercise

A loan applicant has a **default probability of 0.2**. A default costs the bank **€9,000**; rejecting a customer who would have paid costs **€1,000** in lost margin.

Accept or reject?

Speaker notes

Expected cost of accepting: the bank loses €9,000 with probability 0.2, so $0.2 \cdot 9{,}000 = €1{,}800$. Expected cost of rejecting: it forgoes €1,000 with probability 0.8 that the applicant would have repaid, so $0.8 \cdot 1{,}000 = €800$. Rejecting is cheaper by €1,000.

The common intuitive answer, "accept, because 0.2 is low", treats 0.5 as a neutral threshold. It is wrong because the two errors have different costs: a probability of 0.2 is high when the loss it guards against is nine times the alternative. The decision depends on the product of probability and cost, not on the probability alone.

# Mini-Exercise

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Accept or reject?

Accepting costs $0.2 \cdot 9{,}000 = €1{,}800$ in expectation; rejecting costs $0.8 \cdot 1{,}000 = €800$. **Reject —** although $p$ is far below 0.5.

# What the Threshold Tells Us

THE INTUITIVE RULE

**reject if $p > 0.5$**

accepts this applicant — whose default risk is **twice** the bank's break-even

THE RULE THE COSTS IMPLY

**reject if $p > 0.1$**

because a default costs **9×** what a lost customer does: $p^* = \frac{1{,}000}{1{,}000+9{,}000}$

The threshold comes from the **costs**, not from the probability. 0.5 is only the right cut-off when both mistakes are equally expensive — which they almost never are.

Speaker notes

Derivation of the threshold: with $c_A$ the cost of accepting a defaulter and $c_R$ the cost of rejecting a good customer, reject when $p\,c_A > (1-p)\,c_R$, i.e. $p > p^* = c_R/(c_R + c_A)$. Here $p^* = 1{,}000/10{,}000 = 0.1$, and the applicant's 0.2 is twice that. At $p = p^*$ both expected costs are €900 ($0.1 \cdot 9{,}000 = 0.9 \cdot 1{,}000$): the decision is indifferent, and this is where an additional signal has the greatest value.

Sensitivity: if the margin doubles to €2,000, $p^* = 2{,}000/11{,}000 \approx 0.18$ — a lost customer costs more, so more risk is accepted. If the loan is ten times larger (default cost €90,000, margin unchanged), $p^* = 1{,}000/91{,}000 \approx 0.011$. The model output is the same in every case; only the costs change. The threshold 0.5 is optimal exactly when $c_A = c_R$. The rule also assumes calibrated probabilities: if the scores are systematically too low or too high, $p^*$ is applied to the wrong quantity.

# Takeaway

A prediction without a measure of its uncertainty cannot be compared, weighed against consequences, or handed to a human at the right moment. **Modelling uncertainty and reasoning with it** is what turns a prediction into a decision.

All of this assumes the numbers are **calibrated**: a score of 0.55 is right about 55% of the time. Whether a model's output deserves that reading is a question Lecture 04 answers.

# Sources of Uncertainty

**Guiding Question:** The model says 0.55. Is that because the world is noisy, or because the model has not learned enough — and would more data help?

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

Speaker notes

A score of 0.55 instead of 0.99 has two possible causes. Either the outcome genuinely varies among cases with these features — then even a perfect model, trained on unlimited data, would report about 0.55 — or the model has seen too few similar cases to estimate the probability well, and its 0.55 is itself an uncertain estimate.

The third question separates the two. More data of the same kind shrinks the second cause and leaves the first untouched: no amount of data turns a genuinely ambiguous case into a certain one. This is the distinction between reducible and irreducible uncertainty.

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**Is the transaction itself ambiguous** — would a perfect model, with all the data in the world, still be unsure?

**Or has the model simply not seen enough cases like this one?**

**Would collecting more data help — and could *any* amount of data remove the doubt entirely?**

# Two Examples

**EXAMPLE A · AN AMBIGUOUS OBSERVATION**

Two patients with **identical** records — same age, same symptoms, same test results. One recovers, one does not.

✗ Would more patients like these help? Never.

**EXAMPLE B · AN UNFAMILIAR CASE**

A house-price model trained on flats of 40–120 m² is asked about a **600 m² villa**. It has no examples anywhere near it.

✓ Would ten villas in the training data help? Completely.

Which uncertainty can be reduced, and which is part of the problem itself?

Speaker notes

Example A is irreducible: the records are identical, so no model that sees only these records can separate the two patients, and more patients with the same records only estimate the outcome frequency more precisely. A better model does not help either, unless it has access to *more attributes*. The noise is therefore relative to the information available — "inherent" means inherent given the features, not inherent in nature.

Example B is reducible: the model is extrapolating far beyond its training range (40–120 m²), so its prediction for a 600 m² villa rests on the assumed functional form, not on data. Training examples near 600 m² remove most of this uncertainty. A model that reports only a point prediction gives no sign that it is extrapolating.

# Aleatoric vs. Epistemic Uncertainty

| | Aleatoric | Epistemic |
|---|---|---|
| **Name** | *alea*, dice — noise in the world | *episteme*, knowledge — ignorance in the model |
| **Where it comes from** | the outcome genuinely varies given what we observe | we have not seen enough to pin the model down |
| **More data?** | ✗ never — irreducible | ✓ yes — reducible |
| **Our example** | identical patients, different outcomes (A) | the 600 m² villa (B) |

Speaker notes

Aleatoric uncertainty (Latin *alea*, die) is variation of the outcome given the observed inputs; formally, the spread of $p(y \mid x)$ under the true process. Epistemic uncertainty (Greek *epistēmē*, knowledge) is uncertainty about which model or parameters are correct; in a Bayesian model, the spread of the posterior $p(\theta \mid D)$. Example A belongs in the aleatoric column, Example B in the epistemic one.

The distinction is operational, not a matter of the names: a case is classified by asking whether more data of the same kind would reduce the uncertainty. Two caveats apply. Aleatoric uncertainty is relative to the chosen features — adding a feature can turn part of it into explained variation. And epistemic uncertainty shrinks with data only where the data fall: a large training set far from the query point leaves it unchanged.

# Why the Distinction Matters

| If the uncertainty is… | The right response | The expensive mistake |
|---|---|---|
| **aleatoric** (in the problem) | model the noise; set a policy for the ambiguous cases | buying more data that cannot help |
| **epistemic** (in our knowledge) | collect data where the model is thin; defer to a human who knows more | accepting it as inherent and never improving |

A model that can tell the two apart can report that it is **out of its depth**, not merely unsure.

> A probabilistic model can represent **both** kinds of uncertainty. A Bayesian one can even report them separately — noise and ignorance as two terms of one prediction (Lecture 07). That is why the distinction is worth a name.

Speaker notes

The separation follows from the law of total variance over the posterior: $\mathrm{Var}[y \mid x, D] = \mathbb{E}_{\theta \mid D}\big[\mathrm{Var}(y \mid x, \theta)\big] + \mathrm{Var}_{\theta \mid D}\big[\mathbb{E}(y \mid x, \theta)\big]$. The first term is the expected noise (aleatoric), the second the disagreement between plausible parameter values (epistemic). In Bayesian linear regression with noise precision $\beta$ and posterior covariance $S_N$ this becomes $\sigma^2(x) = 1/\beta + \varphi(x)^\top S_N \, \varphi(x)$; the second term shrinks as data accumulate near $x$, the first does not.

A point-estimate model reports at most the first term, or neither. On an unfamiliar input such as the 600 m² villa it can therefore be confidently wrong: nothing in its output signals that it is outside its data.

# Takeaway

> Uncertainty has two sources: **noise in the world** and **gaps in our knowledge**. Only the second shrinks with data, and a decision — collect more, defer, act — depends on knowing which one you face.

Speaker notes

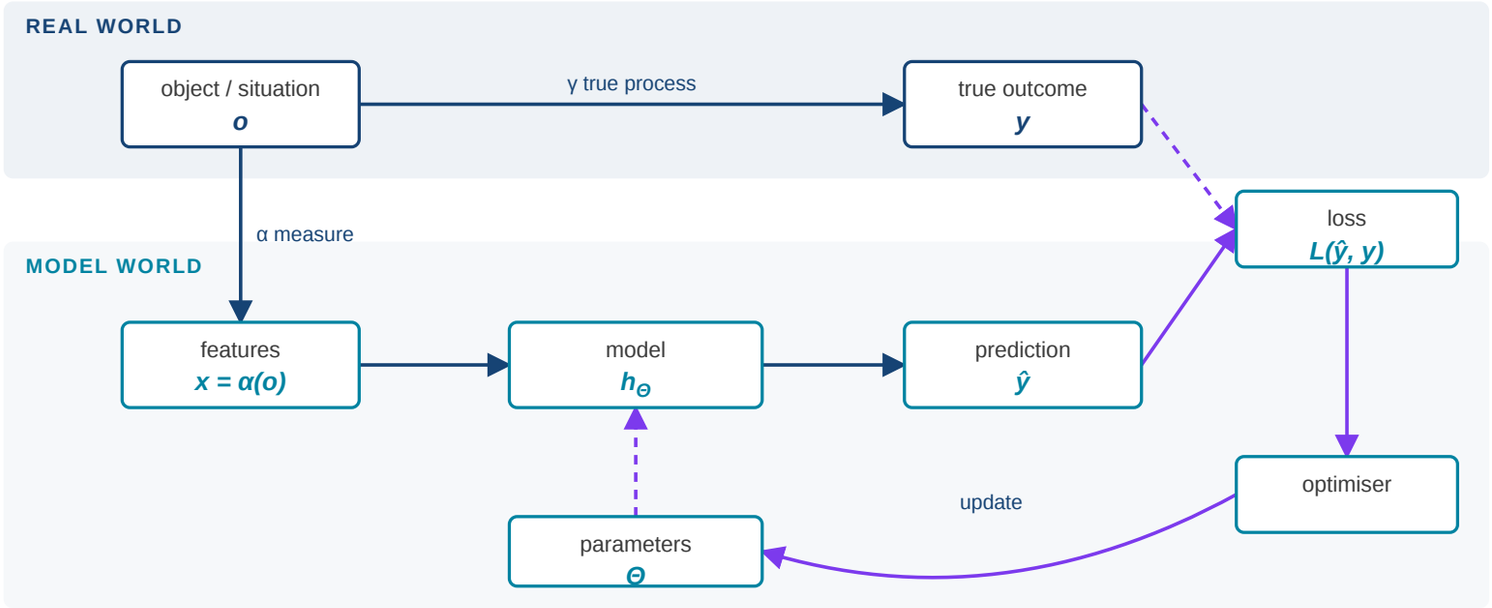
The practical test for any uncertain prediction is whether more data of the same kind would reduce the uncertainty. If yes, the uncertainty is epistemic and collecting data or deferring to someone better informed is worthwhile. If no, it is aleatoric relative to the current features: the remaining options are a decision policy for ambiguous cases or new features. Probability theory provides the means to represent both kinds, combine them and update them; this is developed from Lecture 02 onward.

# Uncertainty in the Learning System

**Guiding Question:** You have built learning systems before. Which of their parts were uncertain — and which of those did your model ever report?

# The Learning System You Already Know

The optimiser changes the parameters — and the training loop closes



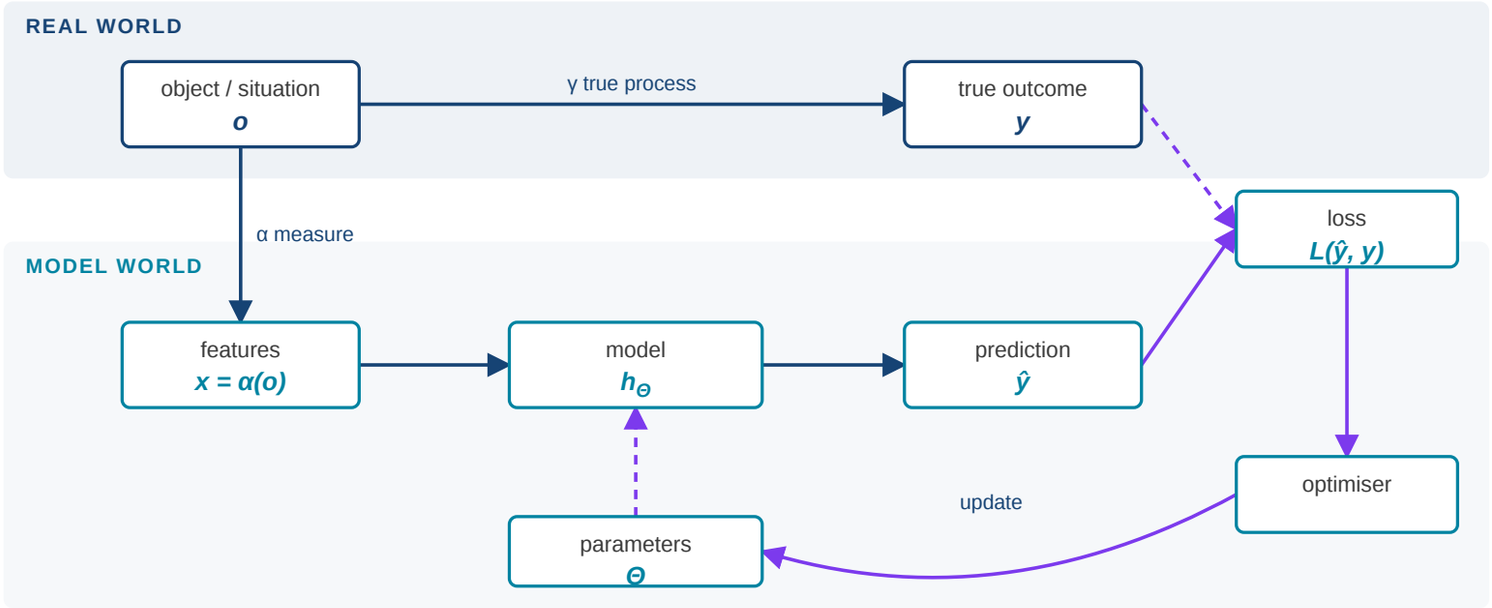
Speaker notes

The figure builds the supervised learning system in four steps. The real world holds a situation $o$ and its true outcome $y$, linked by the unknown true process $\gamma$. Step 1 adds the representation $\alpha$: the features $x = \alpha(o)$ are what is measured, never the situation itself. Step 2 adds the model $h_\Theta$, which maps $x$ to a prediction $\hat{y}$. Step 3 adds the loss $L(\hat{y}, y)$, which compares prediction and outcome. Step 4 adds the optimiser, which updates the parameters $\Theta$ from the loss; the loop repeats until $\Theta$ is fixed. In short: data, model, loss, optimiser.

The question has a factual answer for most standard pipelines: none of these boxes reports its uncertainty. A typical trained classifier returns a label or a score; the closest thing to an uncertainty statement is an aggregate test error, usually reported without an interval although it is estimated from a finite sample. Yet every box is a source of uncertainty, from the noise in $\gamma$ to the finite test set.

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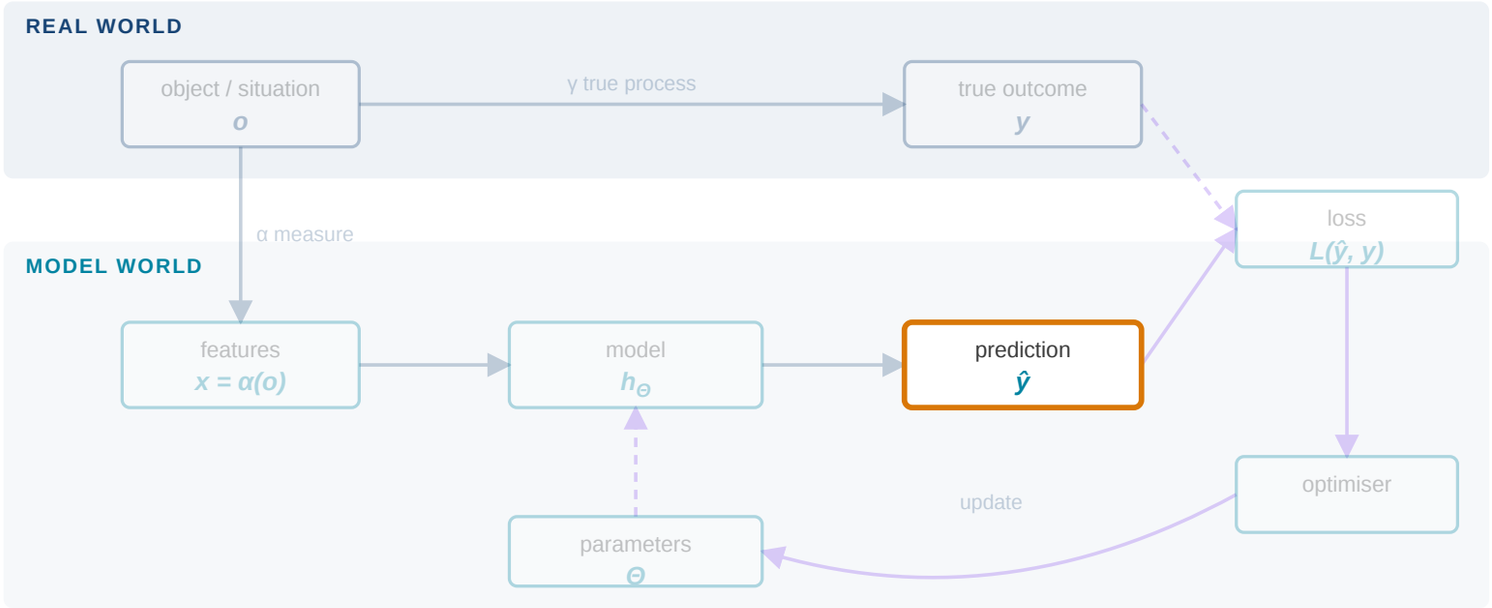
The optimiser changes the parameters — and the training loop closes




Which of these boxes did any model you have trained ever report its uncertainty about?

# The Same System, Read for Uncertainty

In the evaluation




a finite test set, and a future that may not resemble the past

Speaker notes

The learning-system figure, read for uncertainty. Each of the seven steps highlights one box and names the uncertainty located there:

1. **the world** ($o$, $y$) — identical situations, different outcomes; this is aleatoric uncertainty
2. **the data** ($x$) — noisy measurement, missing values, a sample that does not represent the population the model will meet
3. **the model** ($h_\Theta$) — the chosen family may not contain the true relationship, and its assumptions may not hold (misspecification)
4. **the parameters** ($\Theta$) — too few examples to pin them down; this is the epistemic uncertainty that more data reduces
5. **the objective** ($L$) — the loss is a design choice and may not measure what matters for the task
6. **the optimisation** — stochastic steps, random initialisation and approximate convergence give different $\Theta$ from the same data
7. **the evaluation** ($\hat{y}$) — the test set is finite, so every reported metric is an estimate, and deployment data may not resemble the test data

Only steps 1 and 4 match the aleatoric/epistemic pair directly. The others mix both kinds (measurement noise is aleatoric, an unrepresentative sample is not) or concern design choices, and more data of the same kind does not necessarily remove them: a misspecified model stays misspecified, and a biased sample stays biased however large it grows.

# Uncertainty in vs. about the Loss

| | What is uncertain | Kind | More data? |
|---|---|---|---|
| **In the loss** | the targets $y$ are noisy | aleatoric | ✗ never |
| | the training sample is finite | epistemic | ✓ yes |
| **About the loss** | whether this objective represents the task at all | epistemic, but a modelling judgement | ✗ only knowledge of the task |

> Do not say "the loss is random". The loss is an ordinary function — say what is uncertain: the data it is computed from, or whether it is the right objective.

Speaker notes

Inside the loss, the two kinds of uncertainty appear exactly as before. The targets $y$ are noisy, so the loss of a fixed model on a new example is a random variable (aleatoric). The training loss is an average over a finite sample, so it estimates the expected loss with sampling error, and the minimiser fitted to it differs from the one fitted to the population (epistemic; shrinks with more data).

Uncertainty *about* the loss is of a different kind: it concerns whether the objective represents the task at all — for example, whether squared error is appropriate when the costs of over- and under-prediction differ. This is lack of knowledge, so epistemic in the broad sense, but more of the same data does not settle it; it is a modelling judgement, related to model misspecification.

The loss itself is a deterministic function of its arguments. The statement "the loss is random" is imprecise; what is random is the data it is evaluated on.

# Uncertainty in vs. about the Loss

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Do not say "the loss is random". The loss is an ordinary function — say what is uncertain: the data it is computed from, or whether it is the right objective.

Uncertainty is not a property of the final prediction. It **enters throughout the system** — and probabilistic machine learning is the language for representing and reasoning about it at every one of these places.

# Reasoning Under Uncertainty

**Guiding Question:** A proof either holds or it does not. A learned prediction is never a proof — so what kind of conclusion is it, and how strong is it?

# Three Modes of Reasoning

| | From | To | What it guarantees |
|---|---|---|---|
| **Deduction** | premises | what **must** follow | certain — but adds no new content |
| **Induction** | observed patterns | a **regularity**, a hypothesis | nothing — content-adding, not guaranteed |
| **Abduction** | an observation | the **best available explanation** | nothing — only comparable against rivals |

Machine learning lives in the bottom two rows.

Speaker notes

Deduction is truth-preserving: if the premises are true, the conclusion must be, but the conclusion only makes explicit what the premises already contain (it is non-ampliative). Induction generalises from observed cases to a regularity; it adds content, and no finite number of observations guarantees the generalisation (Hume's problem of induction). Abduction, named by C. S. Peirce and also called inference to the best explanation, infers a hypothesis that would explain an observation; its conclusion is only as good as the comparison with rival explanations.

Machine learning performs induction when it fits a model to data and abduction when a model or a hypothesis is selected because it explains the data best. Deduction occurs inside the process — computing a prediction from fitted parameters, proving a convergence theorem — but never as the step from data to model.

# The Cycle, on a Real Case: Uranus

### DEDUCTION

Newton's laws plus the known planets give the orbit Uranus **must** follow.

### INDUCTION

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

### ABDUCTION

Best explanation: an **unseen planet** further out, pulling on Uranus.

Speaker notes

William Herschel discovered Uranus in 1781. Its predicted orbit was computed deductively from Newton's laws and the perturbations of the known planets; by the 1840s the observed positions deviated systematically from these tables, by more than the measurement error. The abduced explanation was an unseen planet beyond Uranus. Urbain Le Verrier (and independently John Couch Adams) computed where such a planet must be. On 23 September 1846 Johann Galle at the Berlin Observatory found Neptune about one degree from Le Verrier's predicted position.

The episode shows the full cycle: induction establishes the anomaly, abduction proposes an explanation, deduction turns the explanation into a checkable prediction, and observation tests it. An abduced explanation gains support only through such a prediction; "an unseen planet" alone explains any deviation.

# The Cycle, on a Real Case: Uranus

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Newton's laws plus the known planets give the orbit Uranus **must** follow.

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Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

**ABDUCTION**

Best explanation: an **unseen planet** further out, pulling on Uranus.

The cycle turns: the abduced planet becomes a new premise. Le Verrier deduces where it must be; in 1846 Galle points a telescope there and finds **Neptune**, about a degree from the prediction.

# An Anomaly Leaves Two Moves

SAVE THE THEORY

## Add a hidden cause

*The bet:* the theory is right; our picture of the world is incomplete.

*If wrong:* we chase something that is not there.

CHANGE THE THEORY

## Replace the premises

*The bet:* the theory fails here, however well it worked elsewhere.

*If wrong:* we throw away two centuries of correct predictions.

1859: Mercury's orbit turns faster than Newton allows — about **43″ per century** by the modern figure. Which move would you make?

Speaker notes

A failed prediction refutes the conjunction of the theory with its auxiliary assumptions (here: that the list of planets is complete), never the theory alone — the Duhem–Quine thesis. Logic therefore cannot decide which part to give up; that choice is a judgement under uncertainty.

Saving the theory is the reasonable first move for Mercury in 1859. Newtonian mechanics had two centuries of accurate predictions behind it, and the same move by the same scientist had just found Neptune. Le Verrier, who reported Mercury's anomalous perihelion advance in 1859, proposed exactly that: an unseen planet, or a ring of bodies, inside Mercury's orbit. His 1859 value was about 38″ per century; Simon Newcomb's 1882 re-analysis gave the 43″ quoted on the slide and still used today.

# Same Move, Opposite Outcomes

| anomaly | the move | what happened |
|---|---|---|
| **Uranus**, 1840s | a hidden planet | **Neptune** found, 1846 ✓ |
| **Mercury**, 1859 | a hidden planet, "Vulcan" (Le Verrier again) | ? |

Speaker notes

The two cases share the scientist, the method and the kind of anomaly, and end oppositely. The hypothetical planet "Vulcan" was searched for during solar eclipses for decades after 1859; several sightings were claimed, and none was confirmed. In November 1915 Einstein derived the 43″ per century from general relativity, with no parameter adjusted to fit it. A theory that predicts a known anomaly without being tuned to it gains far more support than a theory patched to accommodate it, which is why general relativity replaced Newtonian gravity rather than merely amending it.

The lesson is that the quality of a decision and its outcome are different things. Both of Le Verrier's moves were justified by the evidence available when he made them; what distinguished them was evidence that came later.

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> Both decisions were **reasonable when they were made**. What separated them was evidence that did not exist yet.

# Deciding Before the Evidence Is In

- → **Strength of belief.** After two centuries of success, absorbing one anomaly *into* the theory was the rational first move.
- → **Evidence accumulates.** Every failed search made "a hidden planet" less plausible — belief moved step by step, not in one jump.
- → **Replace, don't just refute.** Newton fell when a rival explained *more*, not at the first failure.
- → **Costs decide.** Chasing Vulcan forever and abandoning a good theory too early are both expensive; which move to make depends on what being wrong costs.

> Beliefs of different strength, updated by evidence, acted on under costs — the whole lecture in one historical case. Lecture 03 gives each of these words its mathematics.

Speaker notes

Each bullet has a precise counterpart in probability theory. *Strength of belief* is a prior probability. *Evidence accumulates* is repeated application of Bayes' rule (Lecture 02): in odds form, posterior odds = prior odds × likelihood ratio, so each failed search, being less probable if Vulcan exists than if it does not, lowers the odds by a factor, and many moderate factors compound into a decisive one. *A rival explains more* is model comparison by the probability each model assigns to the data (the marginal likelihood, or Bayes factor between two models). *Costs decide* is the expected-cost rule of the loan example: the action depends on both the probability and the cost of each error.

The replacement criterion matches the philosophy of science: a theory is in practice abandoned when a rival accounts for its successes and its anomalies, not at the first refutation (Lakatos).

# Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

Speaker notes

The three candidate causes are the sources of uncertainty of this lecture: too little data is epistemic uncertainty in the parameters, noise is aleatoric uncertainty in the world, and the wrong model is the model box of the learning system. Collecting more data is the "save the theory" move — the model family is assumed adequate and the sample incomplete; changing the features or the model family is the "change the theory" move.

If too little data were the cause, 500 additional labelled cases from the new country would have reduced the error. That it did not move is strong evidence against this explanation, though not proof: 500 cases could still be too few if fraud in the new country is rare, and the labels themselves could be wrong. To separate the remaining two, a richer model or new features is tried; if the error drops, the model was wrong. If nothing helps and experts with the same information also cannot tell the cases apart, the uncertainty is aleatoric given the available features.

# Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

**Not too little data —** then the error would have dropped. Left: the wrong model (try new features or a richer model) or noise (nothing helps, and experts with the same information cannot tell either). You saved the theory once; now the evidence says to change it.

# Takeaway

> Learning from data never proves; it **bets**. An anomaly can be absorbed or can overturn the model, and which move is right is decided **under uncertainty** — by how strong the belief is, how the evidence accumulates, and what being wrong costs.

# Beyond True or False

| | Classical reasoning | Probabilistic reasoning |
|---|---|---|
| **Conclusions** | follow, or do not | are supported to *degrees* |
| **Evidence** | premises, taken as given | may be partial or noisy |
| **Certainty** | the conclusion is certain | the conclusion may stay uncertain |
| **New information** | can only add premises | can *change* what we believe |

Classical logic asks *whether* a conclusion follows. Probabilistic reasoning also asks **how strongly the evidence supports each possible conclusion**.

Speaker notes

The left column describes deduction, the right column induction and abduction. Classical logic is *monotonic*: adding premises never removes a conclusion that already followed. Probabilistic reasoning is not: a new observation can lower the probability of a hypothesis that was well supported before, as each failed search for Vulcan did.

A learned prediction is never a theorem. It is an inference from evidence, and the quantity of interest is how strongly the evidence supports it. Probability expresses this as a number between 0 and 1, conditional on the evidence: $P(\mathrm{fraud} \mid \mathrm{features})$ is a statement about this transaction given what was observed, not a claim that the transaction is partly fraudulent.

# What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

1. **Represent** — say how much we believe each possibility
2. **Combine** — put several uncertain pieces of information together
3. **Update** — change our beliefs when new evidence arrives
4. **Decide** — act on beliefs that are not certainties

# What the Language Has to Do

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> **Probability provides that language.** For representing, combining and updating there is essentially one consistent set of rules; deciding adds a second ingredient — what the outcomes cost.

# Takeaway

Machine-learning predictions are inferences from evidence, not derivations from premises. **Probability is the calculus of such inferences** — which is why Lecture 02 is about it, and why it will feel like an answer rather than a prerequisite.

Speaker notes

"One consistent set of rules" refers to Cox's theorem (R. T. Cox, 1946; developed by E. T. Jaynes): any system that represents degrees of belief by real numbers and satisfies a few consistency requirements — among them agreement with classical logic in the limit of certainty — must be equivalent to probability theory, with its sum and product rules. Its exact assumptions are still discussed, but the conclusion is the standard justification for probability as the calculus of belief. De Finetti's Dutch-book argument reaches the same rules from betting: beliefs that violate them admit a combination of bets that loses in every outcome.

The theorem covers beliefs only. Acting on beliefs needs preferences over outcomes as well; expected-utility theory (von Neumann and Morgenstern; Savage) shows that consistent choices maximise expected utility. The loan threshold is an instance: the probability came from the model, the threshold from the costs.

# What Probabilistic ML Enables

**Guiding Question:** Which questions about a machine-learning system will you be able to answer after this course that you cannot answer today?

# After This Course, You Will Be Able to Ask — and Answer

### ABOUT A PREDICTION

- ? How certain is it, and should I trust it?
- ? When should the model refuse to decide?
- ? What alternatives are plausible?

### ABOUT THE DATA

- ? When would more data help?
- ? When is uncertainty unavoidable?
- ? What does this data actually justify?

### ABOUT LEARNING

- ? How should belief change when evidence arrives?
- ? How do I combine prior knowledge with observations?
- ? How do I compare competing explanations?

### ABOUT THE METHODS

- ? Why do the common methods work, and under which assumptions?
- ? When should I expect them to fail?
- ? What is my model assuming without telling me?

Speaker notes

The twelve questions are grouped by their object. Questions about a prediction need a calibrated predictive distribution and a decision rule; the reject option ("refuse to decide") is the deferral band of an expected-cost comparison. Questions about the data are the aleatoric/epistemic distinction made quantitative: more data helps exactly where the epistemic part of the predictive uncertainty is large. Questions about learning are answered by Bayes' rule — prior knowledge enters as the prior, observations through the likelihood, and competing explanations are compared by the probability they assign to the data. Questions about the methods are answered by reading common methods as shortcuts of the probabilistic ideal: least squares is maximum likelihood under Gaussian noise, logistic regression under a Bernoulli model, and each inherits the assumptions of its model.

A standard point-prediction pipeline answers none of these questions with confidence; it lacks the quantities they require.

# What Changes Compared to Traditional Machine Learning

**TRADITIONAL ML ASKS**

**What is the prediction?**

**PROBABILISTIC ML ALSO ASKS**

- ? **How certain** is it?
- ? **What alternatives** are plausible?
- ? **Why** are we uncertain?
- ? **How does new evidence** change what we believe?
- ? **What decision** follows from this?
- ? **What can and cannot** be inferred from this information?

# Takeaway

> Probabilistic machine learning does not replace prediction. It **surrounds every prediction with the questions a decision needs answered** — and gives one language in which all of them can be asked.

Speaker notes

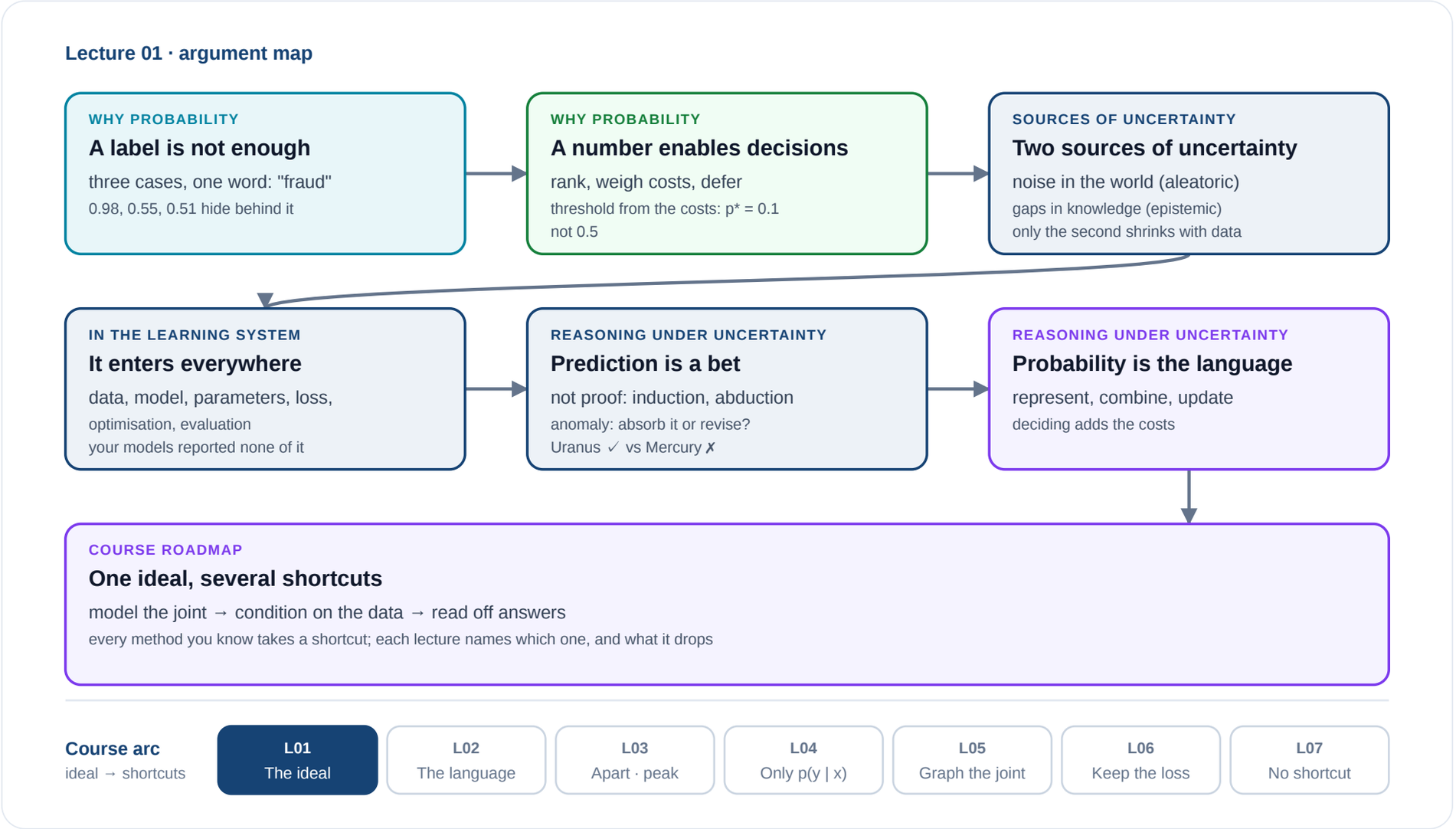
The six questions on the right are the course in brief. *How certain* and *what alternatives* require a full predictive distribution, not a point estimate. *Why uncertain* requires separating aleatoric from epistemic uncertainty. *How new evidence changes belief* is Bayes' rule. *What decision follows* combines probabilities with costs. *What can and cannot be inferred* concerns identifiability and the limits of the data: some questions, such as the direction of a causal effect, are in general not identifiable from observational data without further assumptions, however much of it there is.

Prediction remains the core output; the probabilistic view adds the quantities around it.

# Course Roadmap

**Guiding Question:** Now that the motivation is in place — which lecture answers which of your questions?

# Argument Map



Speaker notes

The map reads top-left to bottom. A label hides what a number shows; the number enables decisions (ranking, thresholds set by costs, deferral); uncertainty has two sources and enters every part of a learning system; a prediction is an inference, not a proof, so it needs a language of degrees; probability is that language.

The bottom strip is the course. This lecture states the ideal — model the joint, condition on the data, read off answers. Each later lecture studies one shortcut that approximates or drops part of the ideal, and Lecture 07 keeps the whole posterior. Every later lecture's argument map carries the same strip with its own row highlighted.

# The Road Ahead

Each lecture answers one of the questions you have just met.

## 02

**Representing uncertainty** *How do we describe uncertainty mathematically?*

Probability, distributions, and how to summarise data.

## 03

**Generative models and naive Bayes** *What does a classifier look like when it is built straight from Bayes — and what does it cost?*

Keeping the peak, modelling the joint $p(x, y)$, naive Bayes.

## 04

**Discriminative models** *What do linear and logistic regression assume — and when is their output a probability?*

Modelling only $p(y \mid x)$: linear and logistic regression, calibration.

## 05

**Structured models** *How do we write down a large, structured set of beliefs?*

Graphical models: drawing the assumptions, fitting them, computing with them.

## 06

**Judging models** *What is the quantity we actually want to minimise — and what did we get instead?*

Evaluation and empirical risk minimisation.

## 07

**Deciding and refining** *How do we keep uncertainty all the way to the decision?*

Mixtures, Bayesian regression, and what comes after.

Speaker notes

The six lectures follow one arc: represent uncertainty (Lecture 02), build a classifier from the joint distribution (Lecture 03), model only the conditional $p(y \mid x)$ (Lecture 04), structure a large joint distribution as a graph (Lecture 05), judge models by their risk (Lecture 06), and keep uncertainty through to the decision (Lecture 07).

03 and 04 form the generative/discriminative pair: a generative model specifies $p(x, y)$ and obtains $p(y \mid x)$ by Bayes' rule; a discriminative model specifies $p(y \mid x)$ directly and cannot answer questions about $x$ alone.

# The Ideal, in Three Moves

Every method in this course is measured against the same three moves, in the language you are about to learn:

### Model the joint

Write down what the world would have to be like to produce what we observed — the data *and* the unknowns together.

### Condition on the data

Inference is conditioning: fix what was observed, and let the rules of probability say what follows for the rest.

### Read off answers

Predictions, uncertainties and decisions are all quantities computed from that one conditioned model.

> Every one of these words — joint, condition, inference — gets a precise meaning in Lecture 02. They appear here only so that you recognise them when they return.

Speaker notes

The three moves are the Bayesian ideal. Write the joint distribution of observed data $D$ and unknowns $\theta$ as $p(D, \theta) = p(D \mid \theta)\, p(\theta)$. Fixing the observed $D$ and renormalising gives the posterior $p(\theta \mid D) = p(D \mid \theta)\, p(\theta) / p(D)$ — inference is conditioning. Every answer is then a quantity computed from that posterior, for example the predictive distribution $p(y \mid x, D) = \int p(y \mid x, \theta)\, p(\theta \mid D)\, d\theta$ or the expected cost of an action.

The ideal is rarely carried out exactly: the normalising constant $p(D)$ and the integral over $\theta$ are usually intractable, and specifying a full joint distribution may be infeasible. The methods in this course are measured by which of the three moves they approximate or omit.

# …and the Shortcuts Everyone Takes

Most methods you already know take a shortcut — and each one drops something.

| Shortcut | Instead | What is lost | L |
|---|---|---|---|
| **Keep the peak** | a single best parameter (maximum likelihood or MAP) | how sure we are about it | 03–04 |
| **Assume it apart** | factorise the joint by independence | the dependences assumed away | 03 |
| **Only what you need** | model $p(y \mid x)$, never the joint (discriminative) | every other question | 04 |
| **Graph the joint** | state exactly which independences hold | cheap inference, in general | 05 |
| **Keep the loss** | minimise average error on the sample (ERM) | every probability | 06 |
| **No shortcut** | keep the whole posterior — EM only for the hidden labels | computation; it cannot see a wrong model family | 07 |

> Knowing which shortcut a method takes tells you which questions it can no longer answer.

Speaker notes

Each row names what a method computes instead of the full posterior and which answers are lost. Maximum likelihood keeps $\hat{\theta} = \arg\max_\theta p(D \mid \theta)$ and discards the spread of $p(\theta \mid D)$, so predictions ignore parameter uncertainty. Strictly, maximum likelihood maximises the likelihood; the most probable parameter under the posterior is the MAP estimate, and the two coincide under a flat prior. Naive Bayes factorises $p(x \mid y) = \prod_j p(x_j \mid y)$; the dependences between features are assumed away. A discriminative model fits $p(y \mid x)$ only, so it cannot generate inputs, handle missing features by marginalisation, or detect unusual $x$. A Bayesian network states exactly which conditional independences hold; exact inference in a general network is NP-hard. Empirical risk minimisation minimises the average loss on the sample and need not produce probabilities at all. The full posterior is the most expensive option and still conditions on a chosen model family; it cannot reveal that the family itself is wrong.

Most familiar methods — linear and logistic regression fitted by maximum likelihood, decision trees, standard neural networks — take one or more of these shortcuts. The generative side is treated in Lecture 03, the discriminative side in Lecture 04.